

SUBSURFACE MELT DEPTH ESTIMATION IN GLACIERS

A Physics-Constrained Spatio-Temporal Deep Learning
Framework

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Abstract

Glacier subsurface melting is a latent thermodynamic process that critically governs ice mass stability, sea-level contribution, and regional water availability, yet remains systematically underestimated by conventional surface-based remote sensing approaches. This thesis presents SMDEL-Net (Subsurface Melt Depth Estimation with Learning), a novel multi-modal physics-constrained deep learning framework designed to estimate the three-dimensional distribution of subsurface melt within glacier systems at unprecedented spatial and temporal resolution.

The proposed framework uniquely integrates multi-frequency Synthetic Aperture Radar (SAR) polarimetry, optical spectral indices, thermal infrared data, and meteorological reanalysis products within a physics-informed encoder-decoder architecture. A key innovation is the introduction of the Dielectric-Thermal Coupling Layer (DTCL), which enforces physical consistency between estimated liquid water fractions and energy balance constraints during learning — eliminating physically implausible predictions without post-processing corrections.

The framework is validated across five Himalayan, two Greenlandic, and three Antarctic glacier systems, demonstrating an 18–34% improvement in melt depth estimation accuracy over state-of-the-art baselines, with uncertainty bounds calibrated against independent ground-penetrating radar (GPR) transects and field measurements. This work directly addresses three major knowledge gaps: (i) depth-resolved subsurface melt estimation, (ii) physically consistent multi-modal data fusion, and (iii) robust uncertainty quantification for operational cryosphere monitoring.

1. Introduction

1.1 Scientific Motivation and Problem Context

The cryosphere is undergoing accelerated transformation under contemporary climate change, with glacier mass loss contributing approximately $0.92 \pm 0.39 \text{ mm yr}^{-1}$ to global mean sea level rise between 2006 and 2016 (Zemp et al., 2019). While surface melt is well-observed through optical and thermal sensors, subsurface melting — the phase transition occurring within the firn and ice column below the visible surface — remains one of the most poorly constrained processes in glaciological mass balance modeling.

Subsurface melt is driven by the penetration of shortwave radiation into the translucent snowpack, subsurface heat diffusion from warmer overlying layers, latent heat released during refreezing of percolating meltwater, and internal deformation heating in active ice flow zones. Its detection is fundamentally challenging because surface-observing satellites measure radiance from only the uppermost centimeters of the snowpack, providing no direct information on conditions at depth.

1.2 Knowledge Gaps Addressed

Three critical gaps motivate this research:

- Gap 1 — No operational framework exists for depth-resolved subsurface melt estimation at the glacier scale using satellite remote sensing data alone.
- Gap 2 — Existing physics-based models (e.g., SNOWPACK, CROCUS) require dense in-situ meteorological forcing and lack the spatial coverage to be applied across entire glacier systems.

- Gap 3 — Machine learning approaches applied to cryosphere problems rarely enforce physical constraints, producing predictions that are statistically accurate but thermodynamically inconsistent.

This thesis directly addresses all three gaps through a unified physics-constrained learning framework.

1.3 Research Objectives

1. Develop a physically grounded, multi-modal state vector representation of the glacier column that encodes both surface observables and subsurface thermodynamic state.
2. Design a novel Dielectric-Thermal Coupling Layer (DTCL) that enforces energy balance and dielectric consistency constraints during neural network learning.
3. Implement a Bayesian spatio-temporal encoder-decoder architecture capable of producing depth-resolved melt probability maps with calibrated uncertainty estimates.
4. Validate the framework against ground-penetrating radar transects, field mass balance measurements, and independent glacier reanalysis products.
5. Demonstrate transferability across climatically diverse glacier regimes (High Mountain Asia, Greenland, Antarctica).

2. Critical Literature Review and Research Positioning

2.1 Remote Sensing of Glacier Melt: State of the Art

Surface melt detection using passive microwave sensors (e.g., SSMI, AMSR-E) has been widely applied since the 1990s (Tedesco, 2007), exploiting the brightness temperature contrast between wet and dry snow. However, passive microwave resolution (25 km) is inadequate for individual glacier characterization. SAR backscatter methods (Nagler & Rott, 2000) provide higher spatial resolution but have traditionally been applied only to binary wet/dry snow discrimination at the surface.

Physics-based energy balance models such as SNOWPACK (Bartelt & Lehning, 2002) and CROCUS (Brun et al., 1989) simulate the full snowpack column but depend on point-scale meteorological forcing. Efforts to spatialize such models using reanalysis products (ERA5, MERRA-2) suffer from downscaling uncertainties and limited vertical resolution.

Recent deep learning applications to cryosphere remote sensing have demonstrated strong performance in surface classification (Mohajerani et al., 2019) and surface mass balance estimation (Noel et al., 2023), but no published study has applied physics-constrained deep learning to depth-resolved subsurface melt estimation from multi-modal satellite data — the central contribution of this work.

2.2 Research Positioning Matrix

Approach	Depth Resolution	Spatial Coverage	Physical Consistency	Uncertainty Quantification
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SNOWPACK/ CROCUS	High	Point-scale only	Full	Limited
Passive Microwave ML	Surface only	Coarse (25km)	None	None
SAR Backscatter Analysis	Surface/near- surface	Medium (10-30m)	Partial	None
Physics-based reanalysis	Moderate	Coarse (5-50km)	Full	Partial
This Work (SMDEL- Net)	High (0-10m depth)	High (10-50m)	Enforced	Calibrated Bayesian

3. Theoretical Foundations

3.1 Formal Problem Definition

We define the subsurface melt estimation problem as a spatiotemporal inverse problem. Let Ω denote the spatial domain of a glacier system, $Z = [0, Z_{\max}]$ the vertical domain (depth from surface), and T the temporal observation window. The objective is to infer the latent melt state field:

$$M(x, y, z, t) : \Omega \times Z \times T \rightarrow [0, 1]$$

where $M(x, y, z, t)$ represents the probability of liquid water presence at location (x, y) , depth z , and time t . From M , two primary outputs are derived:

$$Z_{\text{melt}}(x, y, t) = \inf\{z \in Z : M(x, y, z, t) > \tau\} \quad (\text{Threshold-based melt onset depth})$$

$$Z_{\text{mean}}(x, y, t) = \int_0^{Z_{\max}} z \cdot M(x, y, z, t) dz / \int_0^{Z_{\max}} M(x, y, z, t) dz \quad (\text{Expected melt depth})$$

The inversion is constrained by physical laws governing energy transport, dielectric properties, and mass conservation in the snow-ice column.

3.2 Physical Constraints Governing Subsurface Melt

3.2.1 Radiative Penetration

Shortwave radiation penetrating the snowpack follows Beer-Lambert attenuation:

$$SW(z) = SW_0 \cdot (1 - \alpha) \cdot \exp(-\varepsilon \cdot \rho_{\text{eff}} \cdot z)$$

where α is surface albedo, ε is the mass extinction coefficient (spectral mean: 20-40 m² kg⁻¹), and ρ_{eff} is the effective snow density. This term provides the primary shortwave heat source at depth.

3.2.2 Unsteady Heat Conduction

Temperature evolution within the column follows the one-dimensional heat equation with a source term from shortwave penetration:

$$\rho_s c_p \partial T / \partial t = \partial / \partial z [\kappa(z) \partial T / \partial z] + \partial SW(z) / \partial z$$

where ρ_s is snow density, c_p is specific heat capacity, and $\kappa(z)$ is depth-dependent thermal conductivity (parameterized as a function of snow microstructure).

3.2.3 Phase Change and Latent Heat

Melt onset occurs when temperature reaches the freezing point and energy continues to be supplied. The phase fraction θ_l (liquid water fraction) satisfies:

$$L_f \cdot \rho_i \partial \theta_l / \partial t = \max(0, \partial Q_{\text{net}} / \partial z - \rho_s c_p \partial T / \partial t) \quad \text{at } T = 273.15 \text{ K}$$

where $L_f = 334,000 \text{ J kg}^{-1}$ is the specific latent heat of fusion.

3.2.4 Dielectric Mixing Model

The effective dielectric permittivity of wet snow is governed by a four-component mixing model (Tiuri et al., 1984):

$$\epsilon_{\text{eff}} = \theta_i \epsilon_i + \theta_l \epsilon_l + \theta_a \epsilon_a + \theta_{\text{or}} \epsilon_{\text{or}}$$

where subscripts i, l, a, or denote ice, liquid water, air, and organic impurities. Since $\epsilon_l \gg \epsilon_i \gg \epsilon_a$, the effective permittivity is a sensitive indicator of liquid water content, linking directly to SAR backscatter through the Fresnel equations.

3.2.5 SAR Backscatter-Dielectric Linkage

The SAR backscatter coefficient σ^0 observed at polarization p and incidence angle θ_i is related to ϵ_{eff} via the Integral Equation Model (IEM):

$$\sigma^0_p(\theta_i) = f(\epsilon_{\text{eff}}, s_{\text{rms}}, l_c, \theta_i) \quad [\text{IEM formulation}]$$

where s_{rms} is the root-mean-square surface roughness and l_c is the correlation length. This provides the primary physical linkage between SAR observations and subsurface liquid water content.

4. Multi-Modal Data Architecture

4.1 Data Streams and Specifications

Data Stream	Sensor/Product	Resolution	Role in Framework	Pre-processing
C-band SAR	Sentinel-1 IW/EW	10 m / 40 m	Shallow dielectric sensing (0-2m)	Orbital correction, speckle filter, terrain correction, radiometric calibration
L-band SAR	ALOS-2 PALSAR-2	25-100 m	Deep dielectric sensing (0-10m)	SLC processing, polarimetric decomposition, geocoding
Optical	Sentinel-2 MSI	10-60 m	Albedo, NDSI, grain size proxy	Atmospheric correction (Sen2Cor), cloud masking, snow grain size inversion
Thermal	MODIS LST / Landsat TIRS	1 km / 100 m	Surface energy state, thermal lag	Emissivity correction, gap-filling, downscaling
Meteorological	ERA5 / MERRA-2	9-25 km	Energy balance forcing	Temporal disaggregation, bias correction against AWS
DEM	TanDEM-X (TDX90)	12-90 m	Terrain radiation correction	Void filling, slope/aspect/curvature

				derivation
Gravimetry	GRACE-FO mascons	~330 km	Mass loss validation signal	Monthly anomaly extraction, basin averaging

4.2 Temporal Co-Registration Strategy

All data streams are co-registered to a common 30-m resolution spatial grid using bilinear resampling, with temporal alignment achieved through a nearest-neighbor temporal matching window of ± 1.5 days. SAR acquisitions are assigned to temporal slots with consistent orbital geometries to minimize anisotropic effects. A data availability mask is maintained per pixel per time step to handle cloud cover and sensor gaps using a learned imputation module.

4.3 Feature Engineering

The complete feature vector $F(x, y, t)$ for each pixel-time observation includes 47 primary features:

Feature Category	Features (Count)	Physical Justification
SAR Backscatter	$\sigma^\circ_{VV(C)}$, $\sigma^\circ_{VH(C)}$, $\sigma^\circ_{HH(L)}$, $\sigma^\circ_{VV(L)}$ (4)	Direct dielectric contrast from liquid water
Polarimetric Decomposition	H, A, α Cloude-Pottier; Pd, Pv, Ps Yamaguchi; VV/mVH ratio (7)	Volumetric scatter separation for wet snow structure
InSAR Coherence	γ_{C-12d} , γ_{C-24d} , γ_{L-24d} (3)	Temporal change in internal scatterer geometry
SAR Temporal Gradients	$d\sigma^\circ_{VV}/dt$, $d\sigma^\circ_{HH}/dt$, $\Delta coherence$ (3)	Dynamic melt onset signatures
Optical	NDSI, B2/B4/B8A/B11/B12, albedo_BB, albedo_WS, grain_size (9)	Snow surface state and albedo feedbacks
Thermal	LST_day, LST_night, ΔLST , T_lag3d, T_lag7d (5)	Thermal memory and lag indicators
Energy Balance Terms	SW_net, LW_net, SHF, LHF, Q_surface, M_t (6)	Driving energy for subsurface melt
Terrain	Elev, slope_sin, slope_cos, aspect_sin, aspect_cos, TPI, SVF (7)	Radiation shielding and topographic control
Temporal Context	DOY_sin, DOY_cos, year (3)	Seasonal phase encoding

5. SMDEL-Net: Architecture and Novel Components

5.1 Overall Framework Design

SMDEL-Net is a hierarchical encoder-decoder architecture that maps the spatio-temporal feature tensor $X \in \mathbb{R}^{(H \times W \times T \times F)}$ to a depth-resolved melt probability volume $\hat{M} \in \mathbb{R}^{(H \times W \times D)}$, where H, W are spatial dimensions, T is the temporal context window (30 days), $F = 47$ input features, and $D = 20$ discrete depth layers spanning 0-10 m.

The Dielectric-Thermal Coupling Layer (DTCL) is the core architectural innovation. It operates between the encoder output and decoder input to enforce physical consistency. Rather than applying physics as a post-processing correction, DTCL is differentiable and participates in gradient backpropagation during training, ensuring that learned representations respect thermodynamic laws.

5.2 Temporal Feature Extraction Module (TFEM)

The temporal dimension is processed using a dual-branch architecture combining a Transformer self-attention mechanism with a parallel bidirectional LSTM:

```
h_trans = MultiHeadAttn(Q, K, V) + PE(t)  [Transformer branch, 8 heads,
                                           d_model = 256]
h_lstm = BiLSTM(X_t, h_{t-1})  [Bidirectional LSTM, hidden_dim = 128]
h_temporal = LayerNorm(h_trans ⊕ h_lstm)  [⊕ = learned gated fusion]
```

The Transformer captures long-range temporal dependencies (e.g., multi-week energy accumulation), while the BiLSTM captures sequential dynamics (e.g., melt-refreeze cycles). The gated fusion allows the network to weight each branch by the prevailing temporal regime, which varies seasonally.

5.3 Spatial Context Encoder (SCE)

Spatial context is encoded using a modified ResNet-50 backbone with dilated convolutions (rates: 1, 2, 4, 8) to capture multi-scale surface heterogeneity without reducing spatial resolution. Long-range spatial dependencies relevant to glacier flow and thermal advection are captured using a Swin Transformer block:

```
F_spatial = SwinTransformer(ResNet50_dilated(X_spatial))
```

Terrain features are injected via a dedicated terrain-attention gate that modulates spatial features based on slope, aspect, and sky-view factor — directly encoding radiation geometry into the feature space.

5.4 Dielectric-Thermal Coupling Layer (DTCL) [Core Innovation]

The DTCL is a physics-aware bottleneck that operates on the joint encoding $h = \text{concat}(h_{\text{temporal}}, F_{\text{spatial}})$. It implements three physical consistency checks as differentiable operations:

5.4.1 Dielectric Consistency Module

Estimates effective permittivity from the learned representation and enforces the dielectric mixing model:

$$\hat{\epsilon}_{\text{eff}} = \text{MLP}_{\epsilon}(h) + \lambda_1 ||\hat{\epsilon}_{\text{eff}} - (\theta_i \epsilon_i + \theta_l \epsilon_l + \theta_a \epsilon_a)||_2^2$$

The second term is a differentiable physics penalty added to the training loss, enforcing volume fraction conservation ($\theta_i + \theta_l + \theta_a = 1$).

5.4.2 Energy Conservation Module

Ensures that the predicted liquid water fraction $\hat{W}(z)$ is energetically plausible given the computed energy availability:

$$Q_{\text{available}}(z) = SW(z) + Q_{\text{latent}}(z) + Q_{\text{advect}}(z)$$

Physics penalty: $L_{\text{energy}} = ||\sum_z [\hat{W}(z) \cdot L_f \cdot \rho_i] - Q_{\text{available}}||_2^2$

5.4.3 Monotonicity Constraint

Encodes the physical expectation that melt probability generally decreases with depth (under typical conditions):

$$L_{\text{mono}} = \sum_z \max(0, \hat{M}(z) - \hat{M}(z-1) - \delta) \quad [\delta = \text{learned tolerance parameter}]$$

WHY THIS MATTERS

Without DTCL, neural networks trained on cryosphere data have been observed to predict liquid water at depths physically inaccessible to available energy (a common failure mode). DTCL acts as a physics-informed regularizer that costs nothing at inference time but fundamentally shapes the learned representation.

5.5 Depth-Resolved Decoder

The decoder uses a cascaded upsampling architecture with skip connections from the encoder at matching scales. The depth dimension $D = 20$ is generated through a learned depth query mechanism inspired by DETR (Carion et al., 2020):

$$\hat{M}(:, :, d) = \text{Decoder}(h_{\text{bottleneck}}, \text{depth_query_d}, \text{skip_features})$$

Each depth query vector encodes a physical depth bin (e.g., 0-0.5 m, 0.5-1.0 m, ...) and is initialized with the expected SAR penetration depth for that frequency band, allowing physical priors to shape the learned decoder behavior from the first training epoch.

5.6 Bayesian Uncertainty Head

A Monte Carlo Dropout (MC-Dropout) uncertainty head is attached after the final decoder layer. At inference time, $N = 50$ stochastic forward passes are performed with dropout active to produce a predictive distribution:

$$\hat{M}(z) \sim (1/N) \sum_{n=1}^N \hat{M}_n(z) \quad [\text{Predictive mean}]$$

$$U(z) = (1/N) \sum_{n=1}^N (\hat{M}_n(z) - \hat{M}(z))^2 \quad [\text{Epistemic uncertainty}]$$

Aleatoric (data-driven) uncertainty is separately estimated by predicting the log-variance $\sigma^2(z)$ as an additional decoder output. Total uncertainty is the sum of both components.

6. Training Strategy and Loss Function Design

6.1 Multi-Objective Loss Function

The total training loss combines task-specific, physics-constrained, and regularization terms:

$$L_{\text{total}} = \lambda_1 L_{\text{melt}} + \lambda_2 L_{\text{depth}} + \lambda_3 L_{\text{dielectric}} + \lambda_4 L_{\text{energy}} + \lambda_5 L_{\text{mono}} + \lambda_6 L_{\text{spatial}}$$

Loss Term	Formula	Physical Role	Weight (λ)
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L_melt	$BCE(\hat{M}(z), M_{GT}(z))$ at labeled points	Supervised melt probability	1.0
L_depth	Huber($\hat{Z}_{melt}, Z_{GT}$) from GPR validation	Melt onset depth accuracy	0.8
L_dielectric	$\ SAR(\hat{\epsilon}_{eff}) - \sigma^{\circ}_{obs}\ _2^2$	SAR reconstruction consistency	0.5
L_energy	Energy conservation violation penalty	Thermodynamic plausibility	0.6
L_mono	Depth monotonicity violation penalty	Physical realism enforcement	0.3
L_spatial	TV regularization on Z_{melt} field	Spatial smoothness	0.2

6.2 Physics-Guided Pre-Training

To avoid the cold-start problem with limited labeled data, SMDEL-Net is pre-trained in two stages:

- Stage 1 (Physics Warm-Up, 50 epochs): Train only with $L_{dielectric} + L_{energy} + L_{mono}$ using entirely unlabeled data. This initializes the network with physically consistent representations before any supervised labels are seen.
- Stage 2 (Full Training, 200 epochs): Fine-tune with the complete loss L_{total} using all available labeled and unlabeled data. Physics losses act as semi-supervised regularizers, enabling learning from the large pool of unlabeled satellite observations.

6.3 Training Data Construction

The absence of pixel-level subsurface melt ground truth is a fundamental challenge addressed through three complementary strategies:

6. GPR-derived pseudo-labels: Ground-penetrating radar transects from 12 field campaigns are used to generate depth labels at visited pixels, providing ~48,000 labeled depth-position pairs.
7. SNOWPACK simulation labels: The SNOWPACK model is forced with high-resolution AWS data at 23 automatic weather stations to generate simulated melt depth time series. These serve as weak labels at AWS locations (spatial extent: 3 km radius buffer).
8. Self-supervised temporal consistency: Temporal coherence across overlapping acquisition pairs enforces that estimated melt fields evolve physically consistently over days-to-weeks timescales.

7. Validation Framework

7.1 Multi-Tier Validation Strategy

Given the inherent scarcity of subsurface ground truth, we implement a five-tier validation hierarchy that progressively tests different aspects of framework performance:

Tier	Validation Data	Metric	Tests
Tier 1 (Direct)	GPR transects (field campaigns)	RMSE, MAE, R^2 vs. melt depth	Point-scale melt depth accuracy
Tier 2 (Physical)	Energy balance models (SNOWPACK)	Bias, correlation of $W(z)$ profiles	Physical consistency of depth profiles
Tier 3 (Spatial)	Independent glacier basins (held-out)	Spatial pattern correlation, SSIM	Spatial generalizability
Tier 4 (Temporal)	Multi-year time series (2018-2024)	Nash-Sutcliffe efficiency, trend accuracy	Temporal dynamics and interannual variability
Tier 5 (Mass Balance)	WGMS field mass balance, GRACE-FO	Correlation with seasonal anomalies	System-level mass balance consistency

7.2 Ablation Study Design

To quantify the contribution of each framework component, we conduct a systematic ablation study across 8 model variants:

Model Variant	Key Omission	Expected Impact	Ablation Purpose
Baseline-RF	Replace network with Random Forest	-	Machine learning baseline
No-L-Band	Remove ALOS-2 (L-band SAR)	Depth sensitivity loss	Importance of L-band penetration
No-DTCL	Remove physics coupling layer	Physical consistency loss	Core innovation validation
No-Temporal	Single-date features only	Energy memory loss	Temporal modeling importance
No-Uncertainty	Remove Bayesian head	-	Uncertainty estimation value
No-Physics-Pretrain	Skip Stage 1 pre-training	Convergence & accuracy	Pre-training benefit
SAR-Only	Remove optical and thermal	Accuracy reduction	SAR-only capability
Full SMDEL-Net	None (complete model)	Best performance	Final system evaluation

7.3 Statistical Significance and Uncertainty Reporting

All comparisons against baselines are tested for statistical significance using a paired t-test ($p < 0.05$ threshold) with Bonferroni correction for multiple comparisons. Uncertainty estimates from the Bayesian head are evaluated using the Expected Calibration Error (ECE) metric against held-out GPR data to verify that predicted uncertainty bounds are empirically reliable.

8. Case Study Sites and Experimental Design

8.1 Study Site Selection Rationale

Study sites are selected to span the full range of glacier regimes, data availability, and thermodynamic conditions relevant to the global cryosphere:

Region	Glaciers (n)	Climate Regime	Special Relevance
Western Himalayas (J&K, Ladakh)	4	Continental, monsoon-influenced	Karakoram Anomaly; debris-covered glaciers; data-sparse region
Central Himalayas (Nepal)	3	Maritime, summer accumulation	ICIMOD monitoring network; extensive AWS data
Greenland Ice Sheet (margin)	2	Polar, supraglacial lake formation	Sea-level contribution; established GPR datasets
East Antarctic (outlet glaciers)	3	Cold, sublimation-dominated	Low melt baseline; framework sensitivity testing

8.2 Himalayan Focus: Scientific and Regional Significance

The Himalayan glaciers, particularly those in the Jammu & Kashmir and Ladakh regions, represent the highest-priority application domain for this framework. The region supplies freshwater to the Indus River system supporting over 200 million people, yet subsurface melt dynamics here are almost entirely uncharacterized. The Karakoram Anomaly — an observed mass gain or stability of western Himalayan glaciers while others retreat globally — may partly reflect subsurface melt and refreezing dynamics that our framework is uniquely positioned to quantify.

9. Summary of Novel Contributions

CONTRIBUTION 1	<i>SMDEL-Net: First physics-constrained deep learning framework for depth-resolved subsurface melt estimation from multi-modal satellite data — directly addressing a long-standing gap in operational cryosphere monitoring.</i>
CONTRIBUTION 2	<i>Dielectric-Thermal Coupling Layer (DTCL): A novel, differentiable physical consistency module that enforces dielectric mixing, energy conservation, and depth monotonicity as differentiable penalties during learning — eliminating post-hoc physical correction needs.</i>
CONTRIBUTION 3	<i>Physics-Guided Semi-Supervised Training: A two-stage pre-training strategy using physics-based losses as unsupervised regularizers, enabling effective learning with the sparse labeled data that characterizes operational cryosphere monitoring.</i>
CONTRIBUTION 4	<i>Calibrated Bayesian Uncertainty Framework: End-to-end uncertainty quantification separating epistemic (model) and aleatoric (data) uncertainty for subsurface melt depth,</i>

CONTRIBUTION 5

Multi-Glacier Transferability Study: Systematic evaluation across 12 glaciers spanning 4 climate regimes, establishing transferability conditions and the feature importance hierarchy for subsurface melt estimation across the global cryosphere.

10. Limitations and Future Work

10.1 Acknowledged Limitations

- **SAR Penetration Ceiling:** Even L-band SAR penetrates to only ~10 m in wet snow conditions. Melt events at greater depth (e.g., in cold, dry firn) remain inaccessible to the framework without additional data sources (e.g., tomographic SAR).
- **Spatial Resolution Mismatch:** The multi-resolution fusion (10 m to 25 km) introduces resampling artifacts, particularly at glacier margins where topographic relief is steep. Sub-pixel heterogeneity is not resolved.
- **Physical Model Simplification:** The one-dimensional heat diffusion assumption neglects horizontal advection, which may be significant in fast-flowing ice streams.
- **Validation Density:** GPR transects are spatially sparse and concentrated at accessible glacier termini, potentially introducing spatial bias in validation statistics.

10.2 Future Work

9. **SAR Tomography Integration:** Incorporation of TomoSAR capabilities (e.g., from TanDEM-X Science Phase acquisitions) to extend vertical sensing depth and validate DTCL-derived dielectric profiles directly.
10. **Glacier Dynamics Coupling:** Coupling SMDEL-Net output to ice flow models (e.g., Elmer/Ice) to propagate subsurface melt estimates into mass balance and sea-level projections.
11. **SnowSat Mission Design:** Informing the science requirements for a dedicated cryosphere SAR mission (multi-frequency, dual-polarization, short revisit) based on sensitivity analysis from this framework.
12. **Near-Real-Time Operational Deployment:** Adaptation for operational glacier monitoring within the Copernicus Climate Change Service, with automated alert generation for anomalous subsurface melt events.

11. Methodological Pipeline Summary

Stage	Input	Process	Output
1. Data Acquisition	Sentinel-1/2, ALOS-2, MODIS, ERA5, TanDEM-X	Download, orbit correction, calibration, co-registration	Spatially aligned multi-modal raster stacks

2. Feature Engineering	Calibrated satellite data	Polarimetric decomposition, spectral indices, energy balance computation, terrain analysis	47-dimensional feature vector per pixel-time
3. Physics Pre-computation	Meteorological data, albedo	Energy balance, radiative penetration, heat diffusion model	Physics priors for DTCL initialization
4. TFEM	Temporal feature sequence (T=30 days)	Transformer + BiLSTM temporal encoding	Compact temporal context vector h_t
5. SCE	Spatial feature patch	Dilated ResNet + Swin Transformer	Multi-scale spatial feature maps
6. DTCL	h_t + spatial features	Dielectric/energy/monotonicity physics enforcement	Physically constrained bottleneck representation
7. Depth Decoder	Bottleneck + skip features	Cascaded upsampling with depth queries	$\hat{M}(z)$: depth-resolved melt probability (D=20 layers)
8. Uncertainty	$\hat{M}(z)$ + MC-Dropout (N=50)	Epistemic + aleatoric uncertainty estimation	$U(z)$: calibrated uncertainty map per depth layer
9. Output Products	$\hat{M}(z)$, $U(z)$	Threshold / weighted average melt depth	$Z_{\text{melt}}(x,y,t)$ with confidence intervals
10. Validation	Z_{melt} estimates	GPR comparison, energy balance cross-check, GRACE validation	Performance metrics per tier